

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11: TAXONOMY I (SUB-STRAND 1.1)

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation. It answers the Driving Question: How can we organise the many different living organisms around us so that everyone can identify and study them in the same way? Think back to Lesson 1, when your group sorted the goat, chicken, maize plant, mushroom, algae, bacteria, bean plant and mosquito, and found that other groups sorted them differently and used different Kikuyu, Luo, Kiswahili and English names. Answer each section in your own words in the space under the prompt. Use evidence from the activities you did, including the cards, videos, charts and the Final Classification Challenge. Where a prompt asks you to, give a Claim, Evidence and Reasoning. Write scientific names correctly: genus with a capital letter, species with small letters, and italicised or underlined.

Part 1: Why did our groupings differ, and why is that a problem?

In Lesson 1 you sorted eight familiar organisms by your own rules, and other groups sorted them differently. (a) Describe two different ways groups sorted the organisms. (b) Why could every group say it was correct? (c) Why did this still cause a problem when people tried to identify, study and talk about organisms?

Write your answer here:

Part 2: Why do scientists classify organisms?

Think about the case studies you investigated in Lesson 2. Write a Claim, Evidence and Reasoning explaining why classifying organisms and identifying them correctly matters. Include one reason that links to conserving living things.

Write your answer here:

Part 3: Taxonomic units and scientific names

(a) List the seven taxonomic ranks in order from the broadest to the most specific. (b) Explain what happens to the number of organisms in a group and how similar they are as you move down the ranks. (c) Explain how a scientific name is written and why a two-part scientific name solves the problem of different local names. Use maize (*Zea mays*) as an example.

Write your answer here:

Part 4: Classifying into the five kingdoms (Final Classification Challenge)

Choose any four of these organisms: *Spirogyra*, *Escherichia coli*, *Rhizopus* on stale bread, *Agaricus* (mushroom), *Zea mays*, *Tilapia* (*Oreochromis niloticus*), *Anopheles* mosquito, *Amoeba*, *Acacia*. For each one, name its kingdom (choose from Monera, Protocista, Fungi, Plantae, Animalia) and give the observable evidence you used. Then write one sentence explaining why these characteristics are better for grouping than colour, habitat or usefulness.

Write your answer here:

Part 5: Putting it all together

Compare how your group sorted the eight organisms in Lesson 1 with how you would now classify and name them. Write a final explanation that answers the Driving Question: How can living organisms be systematically organised, named and identified? Your explanation should connect classification, the hierarchy of ranks, scientific names and the five kingdoms, and say why taxonomy is important to people in Kenya.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Correct use of characteristics	Uses cell type, number of cells, cell wall, chlorophyll and mode of nutrition accurately to place organisms in the five kingdoms and explains why they are better than personal criteria.	Uses most characteristics correctly with a few gaps in explanation.	Uses few or incorrect characteristics, or relies on colour, habitat or usefulness.
Evidence and justification (Claim-Evidence-Reasoning)	Gives a clear claim supported by specific evidence from the lessons and reasoning that links the evidence to the claim and to the Lesson 1 puzzle.	Gives a claim with some evidence, but the reasoning is partial.	Gives a claim with little or no evidence or reasoning.
Accuracy of classification and naming	Orders the seven ranks correctly and writes scientific names with the correct format and capitalisation, italics or underlining.	Mostly correct ranks and names with minor format errors.	Several errors in rank order, kingdoms or the format of scientific names.
Communication and collaboration	Writes clearly and logically using correct terms and shows use of group ideas and peer teaching.	Mostly clear, with some terms used loosely.	Unclear or disorganised, with few key terms used.
Local application and conservation connection	Uses Kenyan organisms and places and explains how taxonomy helps identify disease vectors, pests and useful organisms and conserve biodiversity.	Gives a local example or a conservation link, but with limited explanation.	Gives no local example or conservation link.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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